

Drift-Aware Adaptive Aggregation for Federated Learning under Client Data Heterogeneity

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Abstract

Federated learning enables decentralized model training across multiple clients without requiring direct access to their local data. However, realistic deployments frequently exhibit strongly non-IID client data distributions, which can cause local model updates to diverge significantly. Standard aggregation strategies such as Federated Averaging treat all client updates uniformly and may therefore amplify instability under heterogeneous conditions.

This report explores whether *update drift* between client updates and the consensus update direction can serve as a useful signal for server-side aggregation. We introduce Drift-Aware Adaptive Aggregation (DAA), a lightweight aggregation rule that adjusts the influence of client updates based on their deviation from the mean update direction. Experiments on CIFAR-10 under multiple heterogeneous partition regimes demonstrate that update drift increases significantly with client heterogeneity. The proposed aggregation rule exposes this structure and provides a simple mechanism for weighting updates according to their alignment with the global update direction.

1 Introduction

Federated learning (FL) enables collaborative training of machine learning models across multiple decentralized clients while preserving data locality. Instead of sharing raw data, clients train models locally and periodically transmit model updates to a central server, which aggregates these updates to produce a new global model.

One of the most widely used aggregation strategies is **Federated Averaging (FedAvg)**, which averages client updates across participating clients. While effective under IID data distributions, this approach can become unstable when client datasets exhibit significant distributional heterogeneity.

In practical federated deployments, heterogeneity arises naturally due to differences in user behavior, device usage patterns, and geographic context. As a result, client gradients may point in conflicting directions, and uniform averaging may overweight updates that diverge from the consensus direction.

This report explores whether the *geometry of client updates* can provide useful information for improving aggregation robustness. In particular, we investigate whether the distance between each client update and the mean update direction can be used to adaptively weight updates during aggregation.

2 Drift-Aware Adaptive Aggregation

Let w_{global} denote the global model at the beginning of a communication round. After local training, each participating client i returns an updated model w_i .

The update from client i is defined as

$$\Delta_i = w_i - w_{\text{global}} \quad (1)$$

where Δ_i represents the local update vector relative to the current global model. The mean update across K participating clients is

$$\bar{\Delta} = \frac{1}{K} \sum_{i=1}^K \Delta_i \quad (2)$$

Standard aggregation strategies treat all client updates equally or weight them according to dataset size. In contrast, Drift-Aware Adaptive Aggregation (DAA) measures the distance between each client update and the consensus direction.

Specifically, we compute

$$d_i = \|\Delta_i - \bar{\Delta}\|_2 \quad (3)$$

where d_i represents the deviation of client i from the mean update direction. Aggregation weights are then computed as

$$\alpha_i = \frac{\exp(-\beta d_i)}{\sum_{j=1}^K \exp(-\beta d_j)} \quad (4)$$

where β controls the sensitivity of the weighting rule. The updated global model becomes

$$w_{\text{next}} = w_{\text{global}} + \sum_{i=1}^K \alpha_i \Delta_i \quad (5)$$

This introduces a drift-aware weighting mechanism that attenuates highly divergent updates while preserving the standard federated training pipeline.

3 Experimental Setup

Experiments were conducted on the CIFAR-10 image classification dataset using a ResNet architecture.

3.1 Federated Configuration

- Total clients: 20
- Clients sampled per round: 10
- Communication rounds: 20
- Local optimizer: SGD

3.2 Random Seeds

Experiments were repeated with three random seeds:

- 1
- 7
- 42

3.3 Client Partition Regimes

To simulate varying levels of heterogeneity, client datasets were generated using several partition regimes.

Partition	Description
IID	uniform distribution across clients
Dirichlet $\alpha = 0.5$	moderate heterogeneity
Dirichlet $\alpha = 0.3$	strong heterogeneity
Dirichlet $\alpha = 0.1$	extreme heterogeneity

Three aggregation strategies were evaluated:

- FedAvg
- FedProx
- Drift-Aware Adaptive Aggregation (DAA)

4 Results

The experiments reveal several consistent patterns.

First, classification accuracy decreases as client heterogeneity increases. This behavior is consistent with prior observations in federated learning literature.

Second, the distance between individual client updates and the mean update direction increases significantly under stronger heterogeneity regimes. This indicates that update drift can serve as a meaningful indicator of training instability.

Third, the proposed DAA aggregation rule produces interpretable aggregation weights that reflect this drift structure. Updates that deviate strongly from the consensus direction receive reduced influence during aggregation.

5 Discussion

The experiments suggest that client update drift is closely linked to the level of heterogeneity present in the system.

Standard aggregation methods treat client updates as independent contributions. However, under heterogeneous conditions, these updates may vary substantially in both magnitude and direction.

By explicitly measuring the deviation between updates and the consensus direction, the server can gain insight into the stability of the training process and adjust aggregation weights accordingly.

Although the proposed method remains intentionally simple, it highlights the potential value of incorporating update geometry into federated aggregation strategies.

6 Conclusion

This report explored whether update drift between client models and the global model can provide useful signals for aggregation in federated learning.

We introduced Drift-Aware Adaptive Aggregation (DAA), a lightweight rule that weights client updates according to their deviation from the consensus update direction.

Experiments on CIFAR-10 under multiple heterogeneous partition regimes demonstrate that update drift increases significantly with client heterogeneity and can serve as a useful diagnostic signal during federated training.

Future work could explore larger client populations, additional datasets, and theoretical analysis of drift-aware aggregation strategies.